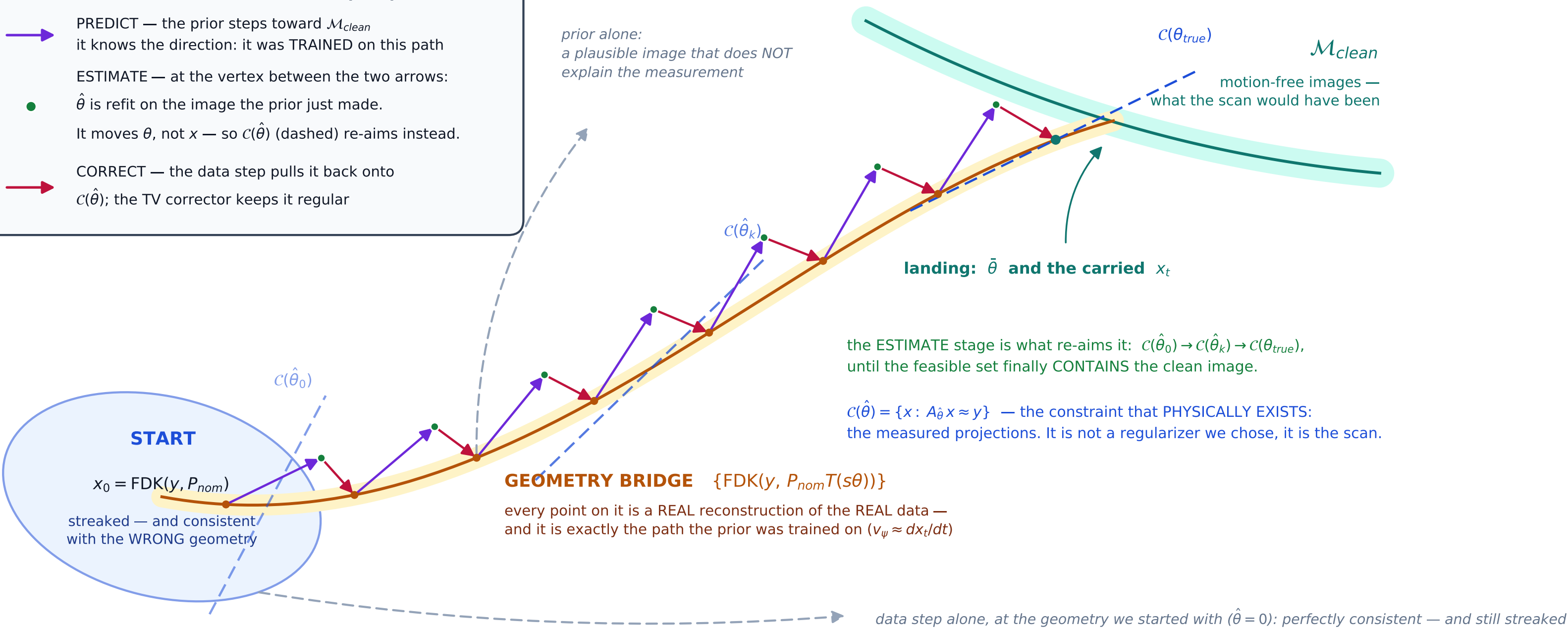


Why the loop works: a learned path to the clean manifold, held on the physics at every step

the prior supplies the DIRECTION (it was trained on this very path); the measured data supplies the CONSTRAINT that keeps the direction honest

ONE STEP OF THE LOOP (x50)

-  **PREDICT** — the prior steps toward $\mathcal{M}_{clean}$
it knows the direction: it was TRAINED on this path
-  **ESTIMATE** — at the vertex between the two arrows:
 $\hat{\theta}$ is refit on the image the prior just made.
It moves θ , not x — so $c(\hat{\theta})$ (dashed) re-aims instead.
-  **CORRECT** — the data step pulls it back onto $c(\hat{\theta})$; the TV corrector keeps it regular



the three moves alternate $N = 50$ times — no single one reaches $\mathcal{M}_{clean}$